LDA or QDA?

SAheart | Prediction, stability & shuffled−label checks

Seed 43207 | 369 training / 93 test | 9 predictors

METHOD

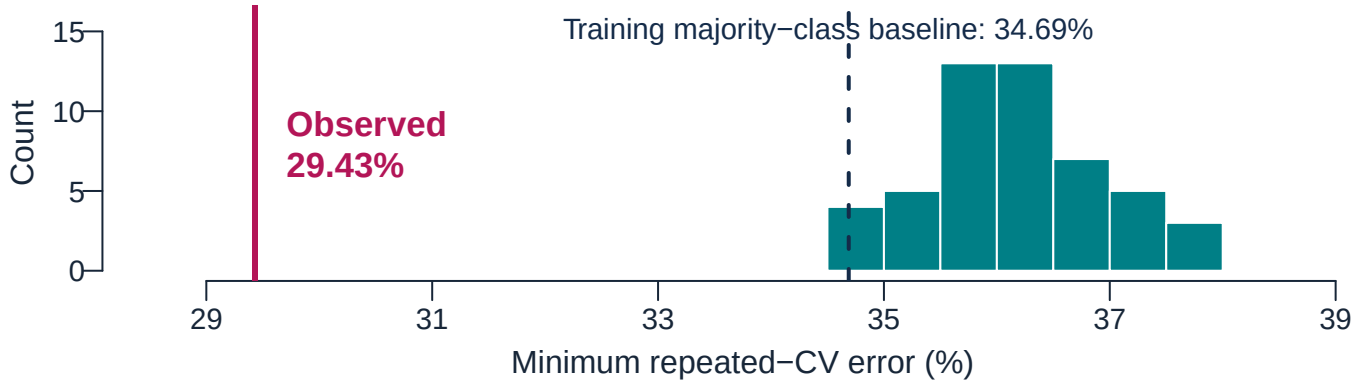
Predict CHD; famhist is coded Absent=1, Present=2. Compare ordinary LDA and QDA with shared stratified 5−fold CV, repeated 10 times. Each fit estimates its own priors, means and covariances; no learned preprocessing. Select the lowest mean error (ties: LDA). Reserve a stratified 20% test set before selection. Shuffle training labels 50 times and repeat the full CV comparison each time.

CROSS−VALIDATION

Model	Mean error	Split SD (pp)
LDA	**29.43%**	**0.63**
QDA	30.84%	1.22

Repeat wins: LDA 8 / QDA 1 / ties 1. Selected: LDA.

PERMUTATION CHECK



Shuffled−label mean: 36.11%. All 50 errors exceeded observed CV error.

TEST ERROR **17.20%**

16 / 93 errors; evaluated once

95% Wilson interval: 10.88%−26.13%

FINDINGS & LIMITATIONS

LDA had lower CV error and less variation across repeated splits; its shared covariance assumption may reduce estimation variance. Final fit: two 9−entry means, one 9 x 9 covariance; priors 0.653 / 0.347. The permutation comparison suggests predictive information, not causality. CV/test differences may reflect small−test sampling variability. Repeated CV folds overlap: split SD is not a confidence interval. Placebo is descriptive; clinical usefulness is unestablished.